IP Addressing Practice

25 questions, easy to hard · Answer key with workings at the back

5 steps: cutting octet (not 255, not 0) → block = 256 − octet → start = ⌊IP octet ÷ block⌋ × block → subnet (then 0s) → broadcast (next start − 1, then 255s). **Hosts** = $2^{(32-n)} - 2$ · **Subnet bits** = n − class bits (A 8, B 16, C 24).

Section 1: Warm-up (binary, classes, masks)

1. Convert to 8-bit binary: (a) 172 (b) 59

2. Convert `11010110` to decimal.

3. Give the class (or special use) of each: (a) 10.4.5.6 (b) 191.255.0.1 (c) 223.1.1.1 (d) 230.0.0.5 (e) 127.0.0.1

4. (a) Write 255.255.252.0 in slash notation. (b) Write /27 as a dotted-decimal mask.

5. How many usable hosts are there in: (a) a /20 (b) a /30 (c) mask 255.255.255.240?

Section 2: Find the subnet, first host, last host and broadcast

6. 192.168.50.137, mask 255.255.255.224

7. 10.20.99.200, mask 255.255.252.0

8. 145.32.180.5, mask 255.255.224.0

9. 60.200.15.70, mask 255.255.0.0 (also give the number of usable hosts)

10. 77.128.200.9, mask 255.192.0.0 *(harder: the cut is in the 2nd octet)*

Section 3: Same subnet?

11. 200.10.5.62 and 200.10.5.65 with mask 255.255.255.192

12. 172.30.17.4 and 172.30.22.250 with mask 255.255.248.0

13. 99.1.64.10 and 99.1.96.10 with mask 255.255.224.0

Section 4: Valid host or not?

14. For each, say **valid host**, **subnet address** or **broadcast**:

a. 192.168.4.63 /26
b. 192.168.4.64 /26
c. 172.16.8.0 /22
d. 172.16.9.0 /22
e. 172.16.11.255 /22
f. 10.0.255.255 /16
g. 10.0.255.255 /8

Section 5: Class + subnetting (Q6 style)

15. IP 201.15.7.99, mask 255.255.255.248. Find: (a) class (b) network ID (c) usable hosts per subnet (d) number of subnets (e) subnet address (f) broadcast

16. IP 18.200.45.3, mask 255.255.255.0. Find: (a) class (b) network ID (c) usable hosts per subnet (d) number of subnets (e) broadcast

Section 6: Design questions (work backwards)

17. What is the **smallest** subnet (the longest mask) that still gives **at least 500** usable hosts? Give it in slash and dotted form.

18. You must split 192.168.10.0/24 into **at least 6** subnets. (a) How many bits must you borrow? (b) What's the new mask? (c) How many hosts per subnet? (d) List the first three subnet addresses.

Section 7: True or false?

19. 255.255.255.250 is a valid subnet mask. **T / F**

20. A /24 network can be split into exactly two /25 networks. **T / F**

21. The broadcast address of 64.12.8.0/21 is 64.12.15.255. **T / F**

22. Under classful addressing, 130.5.0.0 is the network ID of every address that starts with 130.5. **T / F**

23. Borrowing one more bit for subnetting doubles the number of subnets and halves the block size. **T / F**

24. A Class C network has 256 usable hosts. **T / F**

25. An address ending in .0 can never be a valid host. **T / F**

Answer key

Check your working, not just the final number.

Section 1

1. (a) 10101100 (b) 00111011
172 = 128 + 32 + 8 + 4 · 59 = 32 + 16 + 8 + 2 + 1

2. 214 : 128 + 64 + 16 + 4 + 2

3. (a) A (b) B (c) C (d) D (multicast) (e) loopback
The ranges are A 1–126, B 128–191, C 192–223, D 224–239, and 127 is loopback.

4. (a) /22 (b) 255.255.255.224
(a) 252 = `11111100` (6 ones), so 8 + 8 + 6 = 22 · (b) 27 − 24 = 3 ones in the 4th octet, `11100000` = 224

5. (a) 4,094 (b) 2 (c) 14
$2^{12} - 2$ · $2^2 - 2$ · /28 means $2^4 - 2$

Section 2

Q	Block	Subnet	First host	Last host	Broadcast
6	32 (4th)	**192.168.50.128**	.50.129	.50.158	**192.168.50.159**
7	4 (3rd)	**10.20.96.0**	10.20.96.1	10.20.99.254	**10.20.99.255**
8	32 (3rd)	**145.32.160.0**	145.32.160.1	145.32.191.254	**145.32.191.255**
9	none (/16)	**60.200.0.0**	60.200.0.1	60.200.255.254	**60.200.255.255**
10	64 (2nd)	**77.128.0.0**	77.128.0.1	77.191.255.254	**77.191.255.255**

6: 137 is in 128–159 · 7: 99 ÷ 4 = 24.75, so 24 × 4 = 96, range 96–99 · 8: 172 ÷ 32 rounds down to 5, so 160, range 160–191 · 9: the mask is all 255s or 0s, so copy 60.200 and zero the rest. Hosts = $2^{16} - 2$ = **65,534** · 10: the cut is in the 2nd octet, so **both** later octets become 0 (or 255 for the broadcast). 128 is in 128–191.

Section 3

11. No : block 64. 62 is in 0–63 and 65 is in 64–127.

12. Yes : block 8 in the 3rd octet. 17 and 22 are both in 16–23, so both are in subnet 172.30.16.0.

13. No : block 32 in the 3rd octet. 64 is in 64–95 and 96 is in 96–127.

Section 4

14.

a	b	c	d	e	f	g
broadcast	subnet	subnet	**valid ✓**	broadcast	broadcast	**valid ✓**

a/b: /26 means blocks of 64, so .63 ends 0–63 and .64 starts 64–127 · c/d/e: /22 means block 4 in the 3rd octet. The range 8–11 runs from 172.16.8.0 (subnet) to 172.16.11.255 (broadcast), and 172.16.9.0 sits in the middle, so it's valid · f/g: the same address is the broadcast under /16 but a normal host under /8. **The mask decides.**

Section 5

15.

(a) Class	(b) Network ID	(c) Hosts	(d) Subnets	(e) Subnet	(f) Broadcast
C	201.15.7.0	6	32	201.15.7.96	201.15.7.103

201 is in 192–223, so Class C (/24). 248 means /29, so host bits = 3 and $2^3 - 2 = 6$. Subnet bits = 29 − 24 = 5, so $2^5 = 32$. Block 8: 99 ÷ 8 rounds down to 12, and 12 × 8 = 96, giving range 96–103.

16.

(a) Class	(b) Network ID	(c) Hosts	(d) Subnets	(e) Broadcast
A	18.0.0.0	254	65,536	18.200.45.255

18 is in 1–126, so Class A (/8). The mask is /24, so 8 host bits and 254 hosts. Subnet bits = 24 − 8 = 16, so $2^{16} = 65{,}536$.

Section 6

> These work backwards from the formulas. Your slides don't cover this kind of design question, but it's the same maths.

17. /23 = 255.255.254.0

Find the smallest h where $2^h - 2 \geq 500$. h = 8 gives 254 (too few), and h = 9 gives **510** ✓. So n = 32 − 9 = 23. In the 3rd octet that's 7 ones: `11111110` = 254.

18. (a) 3 bits (b) /27 = 255.255.255.224 (c) 30 hosts (d) .0, .32, .64

2 bits gives 4 subnets (too few), and 3 bits gives **8** ✓. Then 24 + 3 = 27. Host bits = 5, so $2^5 - 2 = 30$. Block = 256 − 224 = 32, so the subnets are 192.168.10.0, 192.168.10.32, 192.168.10.64…

Section 7

19. False : 250 = `11111010`. A mask must be all the 1s, then all the 0s, with no gaps. The valid octets are 128, 192, 224, 240, 248, 252, 254 and 255.

20. True : borrowing 1 bit gives $2^1 = 2$ subnets: .0–.127 and .128–.255.

21. True : /21 means block 8 in the 3rd octet. 8 is in 8–15, so the broadcast is 64.12.15.255.

22. True : 130 is Class B, so the network ID is the first 2 octets followed by .0.0.

23. True : subnets go from 2^s to 2^{s+1}, and the block size halves (e.g. 64 → 32). The *usable* hosts are slightly less than half, because you lose 2 per subnet.

24. False : $2^8 - 2 =$ **254**.

25. False : an address ending in .0 is valid when the host bits span more than one octet, e.g. 172.16.9.0/22 (Q14d) or 177.77.205.0/17.